Uncertainty-Aware Open-Vocabulary Gaussian Splatting for Changing Scenes

Anonymous Authors

***Abstract*—A robot that reuses a semantic map must know how much evidence supports each part, and must stop trusting content that has changed. We present a probabilistic open-vocabulary field for 3D Gaussian Splatting that serves both needs. Each Gaussian holds a von Mises–Fisher posterior over the direction of its vision-language feature. Each keyframe counts as at most one observation per Gaussian, because the pixels of one view are strongly correlated. The same sufficient statistics give an epistemic uncertainty, from the weighted number of views, and an aleatoric uncertainty, from a bias-corrected agreement of the views. Little evidence and conflicting evidence thus stay separate. We detect change with a known feature residual and set its threshold per scene, without labels, from the map's own mapping pass. A Gaussian that fails the test is reset. When depth is available, it rejoins the robust map only after new views whose observed depth matches its position. A posterior predictive turns the field into per-pixel class probabilities. We evaluate on Replica and on real rearrangements in RIO10. [TODO: RIO10 access pending: keep RIO10 only if the data arrive.] [TODO: headline result]**

***Index Terms*—Semantic Scene Understanding, Mapping, Probability and Statistical Methods, RGB-D Perception**

I. Introduction

Robots in homes and offices increasingly query their maps with language [1], [2]. Open-vocabulary 3D Gaussian Splatting (3D-GS) [3] supports such queries by attaching vision-language features [4] to the Gaussians of the map [5]–[8]. A robot that reuses the map needs more than the most likely label. It needs to know how much evidence stands behind each Gaussian's semantics, whether that evidence agrees, and whether it is still valid. Indoor scenes change. Objects are moved or removed, and a map that keeps their old semantics answers queries with content that is no longer there.

Several works attach uncertainty to such maps. Bayesian Kernel Inference (BKI) [10], [11] fuses kernel-weighted observations in closed form, and Continuous Semantic Splatting (CSS) [12] brings it to 3D-GS with a Dirichlet posterior over a fixed label set. Other work fuses open-set features on voxels [13], learns an uncertainty [1], [8], or scores the reliability of lifted features [14]. These methods assume a static scene. Change-aware maps [15]–[19] detect moved or removed content by comparing new observations with the map. Two questions fall between these lines of work: how a detected change should enter the uncertainty bookkeeping, and how to set the detection threshold on a new scene without labels.

We answer both questions inside one Bayesian field. Each Gaussian holds a vMF posterior [20] over the direction of its feature. A keyframe contributes one weighted observation per Gaussian: the splat-weighted mean of the pixel features that the Gaussian covers. To detect change, we compare all earlier evidence of a Gaussian with the current view. This statistic is the classic feature residual [16], [19], and we do not claim it as new. Our part is what happens around it: how the threshold is set without labels, which evidence is reset, and when a Gaussian is trusted again. Geometry stays fixed after mapping, so moved content is forgotten at its old place but not inserted at its new one. Fig. 1 shows one keyframe update. Our contributions are:

- **A directional posterior with view-level evidence.** Each Gaussian keeps a vMF posterior, the directional counterpart of Dirichlet semantic mapping, that receives one weighted observation per view. Its epistemic read-out is the weighted number of views. Its aleatoric read-out is a bias-corrected agreement of the views that does not shrink with the amount of evidence.
- **Change handling in the Bayesian bookkeeping.** The threshold of the residual is a quantile of the map's own mapping-pass scores, per history length. This targets a nominal per-view false-alarm rate on static content without labels, and we measure the actual rate. A flagged Gaussian is reset and rejoins the robust map only after a depth check.
- **A posterior predictive.** Per-pixel class probabilities come from a render of each Gaussian's posterior mean feature, so weakly supported Gaussians flatten the prediction. The temperature can also depend on the rendered uncertainties. We test whether this lowers NLL and ECE compared with plug-in features at their own fitted temperature.

We evaluate segmentation, uncertainty ranking and calibration on 8 Replica scenes [21], [22], and stale-content detection on real object rearrangements in RIO10 [23], [24]. **[TODO: RIO10 access pending: keep this clause only if the data arrive, else name the fallback dataset. Add one sentence on the main measured finding, and the anonymous code link or nothing.]**

II. Related Work

Language-embedded 3D-GS. LangSplat [5], Feature 3DGS [6], OpenGaussian [7] and LEGaussians [8] optimize per-Gaussian features through rendering against 2D foundation-model features or masks. LEGaussians also learns

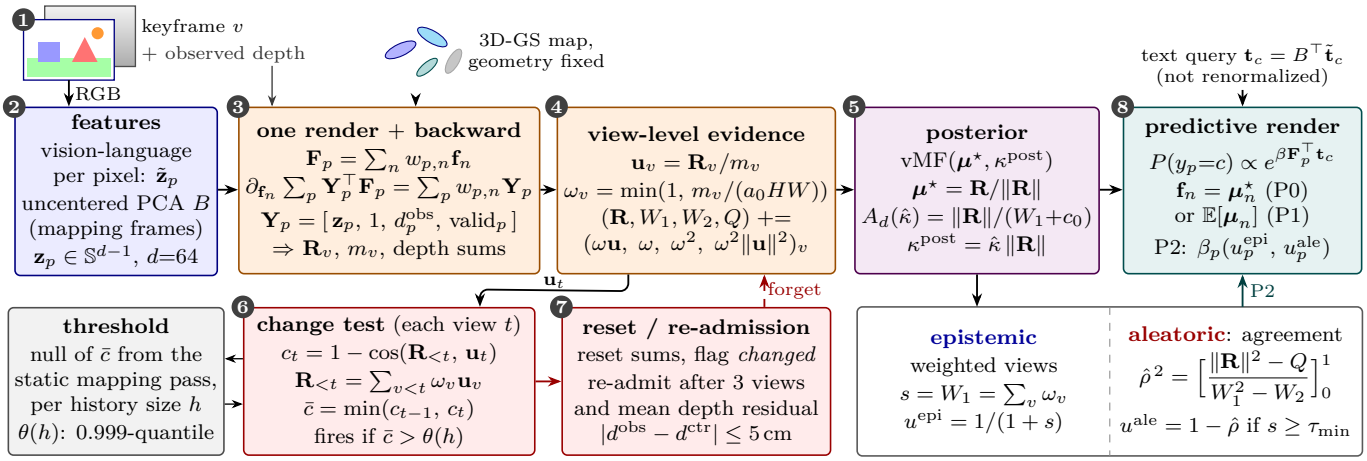


Fig. 1. One keyframe update. (1) A posed keyframe gives RGB and, when available, observed depth. (2) Dense vision-language features, compressed by a PCA fit on the mapping frames. (3) One render and one backward pass [9] give every Gaussian its per-view sums. (4) Each view is one observation $\mathbf{u}_v$ with weight ω_v. (5) Four running sums give the vMF posterior, the support s and the agreement $\hat{\rho}$. (6) A change test with a threshold $\theta(h)$ from the static mapping pass. (7) Reset, and re-admission after a depth check. (8) A predictive render answers text queries. The index n is dropped.

a per-Gaussian uncertainty. Training-free methods [9], [25] give each Gaussian the splat-weighted average of its pixel features, which [9] reads from the gradient of one render. The concurrent NormLift [14] averages per-view mean features. It factors the norm of the un-normalized mean into within-view and cross-view consistency. On static scenes it scores reliability by that norm, shrunk by a Kish count [26] of effective views. We reuse the gradient identity and NormLift's per-view mean. We differ in three ways. Our cross-view agreement is bias-corrected and does not shrink with the amount of evidence. It enters a vMF posterior with a predictive whose calibration we test. We also handle scene change.

Probabilistic semantic mapping. BKI [10] updates a conjugate posterior with kernel-tempered observations. Semantic BKI [11] keeps a Dirichlet posterior over classes in voxels, and E2-BKI [27] adds evidential reasoning and ellipsoidal kernels. CSS [12] brings the Dirichlet model to 3D-GS with the splat weight as the kernel. It separates sparse from conflicting observations over a closed label set fixed at mapping time, and names an open-dictionary conjugate prior as future work. LatentBKI [13] and SLIM-VDB [28] fuse open-set features on voxels under Gaussian models. Bayesian Fields [29] updates per-Gaussian relevance probabilities for tasks given at mapping time. PhysGS [30] splits aleatoric from epistemic variance [31] for physical properties in 3D-GS. We start from the splat-weight kernel of CSS, normalize it per view, and model unit features with a vMF distribution (Sec. III-D).

Directional statistics. The vMF distribution is the standard model for unit vectors [20]. Its mean direction has a conjugate vMF prior [32], and vMF mixtures cluster high-dimensional unit vectors [33]. The implicit language field of LAMP [1] outputs a vMF mean and a concentration, read as aleatoric uncertainty. We train no network. Each Gaussian keeps additive sufficient statistics instead.

Change detection and long-term maps. 3DGS-CD [15] finds rearranged objects from render-image differences and segmentation masks. CL-Splats [16] compares DINOv2 features of renders and images, then re-optimizes only changed regions. GaussianUpdate [17] uses continual learning with generative replay, and VG-Mapping [18] adds and removes Gaussians in changed regions of semi-static scenes. MIF [19] updates its scene memory where the discrepancy of matched scene-graph nodes, a position term plus a feature-cosine term, exceeds a fixed threshold. Chu and Kim [34] let a language model pick stable objects to localize in changing indoor scenes. Our change score is the same kind of feature residual as that of CL-Splats and the semantic term of MIF. It is one minus the cosine between all earlier evidence of a Gaussian and the current view, and a change must persist over two views. We differ in the threshold and the response. The threshold is a quantile of the map's own mapping-pass scores per history size, so it needs no labels. A detection resets the Gaussian's semantic evidence, and new views with a depth check gate its return. Unlike CL-Splats, GaussianUpdate and VG-Mapping, we keep geometry fixed.

III. Method

Each keyframe gives every visible Gaussian one view-level observation, and a few running sums hold its semantic state (Fig. 1, Alg. 1).

A. Preliminaries

The map holds N Gaussians [3], and its geometry is fixed after mapping. At pixel p, Gaussian n has opacity $\alpha_{p,n}$: its base opacity times its projected 2D Gaussian evaluated at p. With the Gaussians sorted by depth, the splat weight is $w_{p,n} = T_{p,n}\,\alpha_{p,n}$ with the transmittance $T_{p,n} = \prod_{j<n}(1-\alpha_{p,j})$. It is the share of pixel p that Gaussian n explains. A per-Gaussian buffer $\mathbf{f}_n$ renders to $\mathbf{F}_p = \sum_n w_{p,n}\mathbf{f}_n$, and $\alpha_p = \sum_n w_{p,n} \le 1$ is the rendered opacity. As in CSS [12], $w_{p,n}$ is the BKI kernel [10], [11].

B. Features and Query Space

A dense extractor, LSeg [35], gives unit pixel features $\tilde{\mathbf{z}}_p$ that are cosine-comparable with unit CLIP text embeddings $\tilde{\mathbf{t}}_c$ [4]. An uncentered PCA with orthonormal basis B, fit on mapping frames only (no text, no labels), compresses them to $d = 64$: $\mathbf{z}_p = B^\top \tilde{\mathbf{z}}_p / \|B^\top \tilde{\mathbf{z}}_p\| \in \mathbb{S}^{d-1}$. Texts are projected without renormalization, $\mathbf{t}_c = B^\top \tilde{\mathbf{t}}_c$, so $\boldsymbol{\mu}^\top \mathbf{t}_c = (B\boldsymbol{\mu})^\top \tilde{\mathbf{t}}_c$ is the full-space cosine between the decoded direction $B\boldsymbol{\mu}$ and the text. Renormalizing $\mathbf{t}_c$ would inflate texts with little energy in the image subspace and can change the argmax label.

C. View-Level Evidence

Keyframe v gives every Gaussian n the sums $\mathbf{R}_{v,n} = \sum_p w_{p,n}\mathbf{z}_p$ and $m_{v,n} = \sum_p w_{p,n}$. We read them from one render and one backward pass. With geometry and opacity detached, $\mathbf{F}_p$ is linear in the buffer, so the gradient of $\sum_p \mathbf{Y}_p^\top \mathbf{F}_p$ with respect to $\mathbf{f}_n$ is $\sum_p w_{p,n}\mathbf{Y}_p$ for any per-pixel target $\mathbf{Y}_p$. We take this identity from Gradient-Weighted Feature Back-Projection [9], and stack $\mathbf{z}_p$, a ones channel and the observed depth with its validity mask into one target. This gives all per-Gaussian sums of Alg. 1, with no per-Gaussian pixel lists. It needs a rasterizer affine in per-Gaussian features. We use gsplat [36] with the buffer in place of the colors and no spherical harmonics.

Pixels of one view are strongly correlated. For example, one ViT-L/16 patch of LSeg covers 16×16 pixels. A pixel count thus overstates the evidence and grows with the image resolution, so we count views. View v gives Gaussian n one observation, the view mean $\mathbf{u}_{v,n} = \mathbf{R}_{v,n}/m_{v,n}$, as in NormLift [14]. Its length $\|\mathbf{u}_{v,n}\| \le 1$ is the coherence inside the view. Its weight $\omega_{v,n} = \min(1, m_{v,n}/(a_0 HW))$ (default $a_0 = 10^{-4}$) does not depend on the resolution, and a view with a smaller footprint counts fractionally. Gaussian n keeps four sums over its views, written without the indices on the right:

$$\mathbf{R}_n = \sum \omega \mathbf{u},\ W_1 = \sum \omega,\ W_2 = \sum \omega^2,\ Q = \sum \omega^2 \|\mathbf{u}\|^2.$$

D. Direction Posterior and Concentration

We use a vMF likelihood $p(\mathbf{z} \mid \boldsymbol{\mu}, \kappa) = C_d(\kappa)\, e^{\kappa \boldsymbol{\mu}^\top \mathbf{z}}$ on $\mathbb{S}^{d-1}$ [20], [33]. Following BKI, we temper the likelihood of pixel p in view v by the kernel $\omega_{v,n} w_{p,n}/m_{v,n}$. A view then carries mass $\omega_{v,n}$ and log-likelihood $\omega_{v,n}\big(\log C_d(\kappa) + \kappa \boldsymbol{\mu}^\top \mathbf{u}_{v,n}\big)$. Given κ_n, the direction posterior is a vMF [32] with natural parameter $\boldsymbol{\eta}_n = \kappa_n \mathbf{R}_n + \kappa_0 \boldsymbol{\mu}_0$, mean direction $\boldsymbol{\mu}_n^\star = \boldsymbol{\eta}_n / \|\boldsymbol{\eta}_n\|$ and concentration $\kappa_n^{\text{post}} = \|\boldsymbol{\eta}_n\|$. The optional text prior is off ($\kappa_0 = 0$), so $\boldsymbol{\mu}_n^\star = \mathbf{R}_n/\|\mathbf{R}_n\|$. For κ_n we use the joint conjugate prior $p(\boldsymbol{\mu}, \kappa) \propto C_d(\kappa)^{c_0} e^{\kappa \mathbf{m}_0^\top \boldsymbol{\mu}}$ with $\mathbf{m}_0 = \mathbf{0}$ (default $c_0 = 1$). The joint posterior is proportional to $C_d(\kappa)^{W_1 + c_0} e^{\kappa \boldsymbol{\mu}^\top \mathbf{R}_n}$, and its MAP concentration solves

$$A_d(\hat{\kappa}_n) = \frac{\|\mathbf{R}_n\|}{W_1 + c_0}, \qquad A_d(\kappa) = \frac{I_{d/2}(\kappa)}{I_{d/2-1}(\kappa)}, \tag{1}$$

where $A_d(\kappa) = \mathbb{E}[\boldsymbol{\mu}^\top \mathbf{z}]$ is the mean cosine of a vMF. We set $\kappa_n = \hat{\kappa}_n$, so $\kappa_n^{\text{post}} = \hat{\kappa}_n \|\mathbf{R}_n\|$. The concentration never changes $\boldsymbol{\mu}_n^\star$ or the label. It only sets how sharp the posterior is. With one-hot features, $\mathbf{R}_n$ equals the count increment of categorical BKI run with our view-level kernel. (CSS [12] uses the pixel kernel $w_{p,n}$, so its counts differ.) The two models share the sufficient statistic and the MAP label but not the distribution, so the vMF model is the directional counterpart of Dirichlet semantic mapping.

E. Epistemic and Aleatoric Uncertainty

Epistemic. The support $s_n = W_1$ is the number of views, each weighted by $\omega \le 1$, and $u_n^{\text{epi}} = 1/(1 + s_n)$. Because $\omega \le 1$, the Kish effective number of views [26] $n_K = W_1^2/W_2$ is never below W_1. It ignores the scale of the weights, so it would count ten small partial views as much as ten full ones.

Aleatoric. Let $\rho_n = \|\mathbb{E}\,\mathbf{u}_{v,n}\|$. It is one when every view shows the same unit feature and near zero when the views disagree. We estimate it by

$$\hat{\rho}_n^2 = \left[\frac{\|\mathbf{R}_n\|^2 - Q}{W_1^2 - W_2}\right]_0^1, \qquad u_n^{\text{ale}} = 1 - \hat{\rho}_n, \tag{2}$$

where $[\cdot]_0^1$ clamps to $[0, 1]$. For independent views with a common mean, the expected squared norm is $\mathbb{E}\|\mathbf{R}_n\|^2 = (W_1^2 - W_2)\rho_n^2 + \sum_v \omega^2\, \mathbb{E}\|\mathbf{u}_{v,n}\|^2$. The ratio inside the clamp is thus unbiased for ρ_n^2 for two or more views. The clamp and the square root leave a small bias at few views. The aleatoric uncertainty is defined once $s_n \ge \tau_{\min}$ (default 3 views). Two simpler estimates mix the amount of evidence into the dispersion. The mean resultant length $\bar{r}_n = \|\mathbf{R}_n\|/W_1$ is biased upward, by up to about $1/\sqrt{n_K}$. A prior-shrunk agreement such as $\|\mathbf{R}_n\|/(W_1 + c_0)$ is small whenever evidence is scarce. NormLift's reliability, its feature norm times $N_{\text{eff}}/(N_{\text{eff}} + 1)$ [14], has this form. Estimator (2) removes the leading bias of $\bar{r}_n$ and does not shrink, so little evidence and conflicting evidence stay separate (Fig. 2). The shrunk form suits $\hat{\kappa}_n$ in (1), where scarce evidence should broaden the posterior.

F. Change Detection, Reset and Re-admission

We test each Gaussian at each view with a family of two-bucket statistics. An older bucket $\mathbf{R}_n^{\text{old}}$ and a recent bucket $\mathbf{R}_n^{\text{new}}$ split $\mathbf{R}_n$ without overlap. Each view moves a share $1 - \lambda^\omega$ of the recent bucket, and of its mass, into the older one. It then adds $\omega_{v,n}\mathbf{u}_{v,n}$ to the recent one (retention $\lambda \in [0, 1)$ per full view). The score $c_n = 1 - \cos(\mathbf{R}_n^{\text{old}}, \mathbf{R}_n^{\text{new}})$ is defined once the support reaches $\tau_{\min}$ and each bucket holds one full view. A change must persist, so the statistic $\bar{c}_n$ is the minimum of the last k defined scores.

On a tuning sequence not used for testing, the 7-Scenes chess scene [37] with simulated object swaps, we chose $\lambda = 0$ and $k = 2$. Then $\bar{c}_n$ is the minimum over the last two views t of $1 - \cos\big(\sum_{v<t} \omega_{v,n}\mathbf{u}_{v,n},\ \mathbf{u}_{t,n}\big)$, the cosine distance between all earlier evidence and the current view. This member is the classic feature-residual test [16], [19]. We do not claim the statistic, only how its threshold is set and the bookkeeping around it. Centering the features on the scene mean hurt in tuning, so it is off.

Threshold from the mapping pass. In tuning, a fixed threshold on $\bar{c}_n$ did not transfer between scenes (fixed-threshold row of Table III). We therefore set it, without labels,

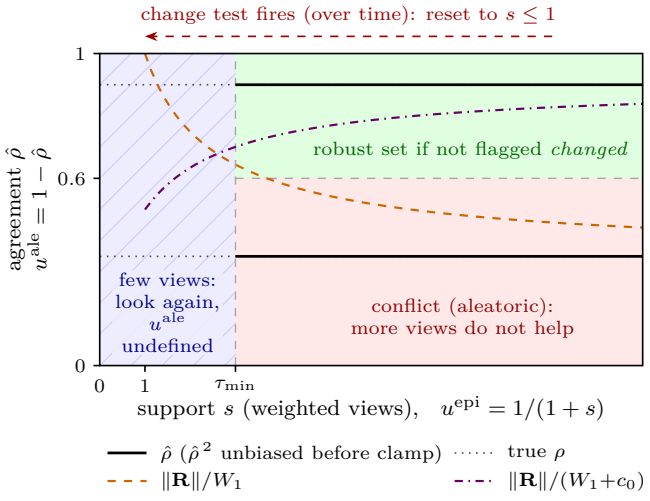


Fig. 2. Support s (weighted views) against agreement $\hat{\rho}$ of one Gaussian. The curves are root-mean-square values for two Gaussians with true agreement ρ (dotted). The mean length $\|\mathbf{R}\|/W_1$ makes the conflicting one look consistent at few views, and the shrunk $\|\mathbf{R}\|/(W_1 + c_0)$ makes the consistent one look conflicting. Our $\hat{\rho}$ is drawn at ρ: before the clamp $\hat{\rho}^2$ is unbiased, so $\hat{\rho}$ stays close to ρ. A fired change test resets s to at most one view.

from the map's own mapping pass. That sequence is assumed static, so its scores sample the no-change null (no resets happen during mapping). Short histories give noisier scores, so we keep one null histogram per history size, the mass $h_n = W_n^{\text{old}}$ of the older bucket, with bin edges at 4, 6, 10, 20 and 40 views. The threshold $\theta(h)$ is the q-quantile of bin h (default $q = 0.999$), a nominal per-view false-alarm rate of 10^{-3} on static content. A bin with fewer than 200 samples borrows the nearest bin with enough. In tuning, a null of per-Gaussian maxima over the mapping pass was inflated by the short early histories.

Reset and re-admission. When $\bar{c}_n > \theta(h_n)$, Gaussian n forgets. Its four sums, buckets and depth record restart from the current view, and it is flagged *changed*. It is re-admitted at the first view at which its view weight since the reset, the view that fired included, is at least 3 and the view-weighted mean of $|d_{v,n}^{\text{obs}} - d_{v,n}^{\text{ctr}}|$ over these views is at most 5 cm. Here $d_{v,n}^{\text{obs}}$ is the observed depth (sensor or estimator, never rendered from the map) averaged over the footprint with the weights $w_{p,n}$, and $d_{v,n}^{\text{ctr}}$ is the camera depth of the Gaussian center. The Gaussians of a removed object see the surface behind them. They stay out while that surface lies more than 5 cm from their centers. A Gaussian with no valid depth since its reset stays out. Without any depth stream, the views alone decide. The *robust set* holds the Gaussians with $s_n \ge \tau_{\min}$, $\hat{\rho}_n \ge 0.6$ and no change flag.

G. Predictive Distribution

For pixel p of a novel view, the class probabilities are

$$P(y_p = c) = \text{softmax}_c\left(\beta\, \mathbf{F}_p^\top \mathbf{t}_c\right), \qquad \mathbf{F}_p = \textstyle\sum_n w_{p,n} \mathbf{f}_n.$$

The render $\mathbf{F}_p$ is not divided by α_p, so low opacity flattens the distribution. *P0* (plug-in) renders $\mathbf{f}_n = \boldsymbol{\mu}_n^\star$. *P1* (posterior) renders the vMF mean $\mathbb{E}[\boldsymbol{\mu}_n] = A_d(\kappa_n^{\text{post}})\, \boldsymbol{\mu}_n^\star$, a mean-field

Algorithm 1 Update of the field with keyframe v

1: Render and backward pass: $\mathbf{R}_{v,n}$, $m_{v,n}$, depth sums
2: **for all** Gaussians n with $m_{v,n} > 0$, in parallel **do**
3: $\quad \mathbf{u} \leftarrow \mathbf{R}_{v,n}/m_{v,n}$, $\omega \leftarrow \min(1, m_{v,n}/(a_0 HW))$
4: $\quad \mathbf{R}_n^{\text{old}} \mathrel{+}= (1-\lambda^\omega)\mathbf{R}_n^{\text{new}}$, $\mathbf{R}_n^{\text{new}} \leftarrow \lambda^\omega \mathbf{R}_n^{\text{new}} + \omega \mathbf{u}$
5: $\quad$ If defined, $\bar{c}_n \leftarrow$ minimum of the last k scores c_n. While mapping, add $\bar{c}_n$ to the null of bin h_n. Later, reset and flag n if $\bar{c}_n > \theta(h_n)$.
6: $\quad \mathbf{R}_n \mathrel{+}= \omega \mathbf{u}$, $W_1 \mathrel{+}= \omega$, $W_2 \mathrel{+}= \omega^2$, $Q \mathrel{+}= \omega^2 \|\mathbf{u}\|^2$
7: $\quad$ Update the depth residual, unflag n if re-admitted
8: **end for**
9: After mapping: $\theta(h) \leftarrow q$-quantile of the null of bin h

treatment of the per-Gaussian posteriors, so weakly supported Gaussians pull $\mathbf{F}_p$ toward zero. *P2* (adaptive) keeps the P0 render with a per-pixel temperature $\beta_p = \beta_0 \exp\big(-\gamma_e u_p^{\text{epi}} - \gamma_a u_p^{\text{ale}} - \gamma_u \mathbf{1}[u_p^{\text{ale}} \text{ undefined}]\big)$. Here u_p^{epi} and u_p^{ale} average u_n^{epi} and u_n^{ale} with the weights $w_{p,n}$ over the Gaussians where they are defined. The pixel value u_p^{ale} is undefined when more than half of α_p comes from Gaussians with undefined u_n^{ale}. The term γ_u gives such pixels their own offset instead of the most confident value $u^{\text{ale}} = 0$. We fit β (P0, P1), $(\beta_0, \gamma_e, \gamma_a, \gamma_u)$ (P2) and the temperature of every baseline by NLL on the validation split.

IV. Experiments

We ask four questions. Is the field as accurate as simple averaging, and does its predictive lower NLL and ECE (Table I)? Does each uncertainty rank the error it is meant for (Fig. 3)? On real rearrangements, does the change test find stale content (Table II)? Which parts matter, and does the reset keep stale content out of the robust set (Table III)?

A. Setup

Replica. We use the Replica scenes [21] as rendered by Semantic-NeRF [22]: 8 scenes with 101 class ids, where id 0 marks unlabeled pixels that we ignore. We validate on office_0 and room_2 and test on room_0, room_1 and office_1 to office_4. Each method maps on every 5th frame of Sequence_1 (180 frames). We evaluate on every 10th frame of Sequence_2 (90 frames), which no method sees while mapping. The query vocabulary of a scene is the set of classes present in its evaluation frames. One ensemble of five prompt templates, chosen on the validation scenes, serves all scenes.

RIO10. RIO10 [23] holds real rescans of 10 rooms in which objects were moved or removed between visits. **[TODO: RIO10 access is pending. This block runs once the data arrive.]** We validate on scenes 01, 04 and 06 and test on the other seven. Each method maps on seqXX_01 and is then updated with seqXX_02 **[TODO: frame strides and pose source]**. Stale ground truth comes from the object annotations [23], [24], which no method reads **[TODO: check that instance ids match across the two rescans, else use the 3RScan annotations]**. An object is stale if it was removed, moved by more than XX.X cm or turned by more than XX.X°. A Gaussian is stale if its nearest annotated point in seqXX_01,

within XX.X cm, belongs to a stale object. We fix these thresholds before we look at any method output. Gaussians that no update view observes, or that the new surface hides in every update view, cannot be judged. We leave them out of the change metrics and report their share. The observed depth for re-admission comes from **[TODO: sensor depth or a depth estimator]**, never from the mesh. The mesh depth of each rescan **[TODO: verify]** shows the new state of the scene, and we use it only in an oracle row.

Map and features. For each scene we fit one 3D-GS map on the mapping frames with gsplat [36], using degree-0 color, XX.X iterations and at most XX.X Gaussians. Every method uses this geometry, which stays fixed afterwards, and the same LSeg features, compressed to $d = 64$ (Sec. III-B). The coverage floor $\tau_{\text{cov}} = XX.X$ and all temperatures are set on the validation scenes. We freeze every choice before the test scenes run, and we run them once. We report the mean and standard deviation over the test scenes ($N = 6$ for Replica, $N = 7$ for RIO10).

Segmentation baselines. All baselines share our geometry, features and splat weights, and each gets its own temperature, fit by NLL on the validation scenes. *2D features* apply the extractor to the evaluation frame itself, with no map. *Averaging* is the pixel-weighted mean feature of Gradient-Weighted Feature Back-Projection [9], with no uncertainty of its own **[TODO: compare with the authors' code on one validation scene]**. *NormLift* [14] keeps the averaging feature. It counts Kish effective views N_{eff} over the per-view pixel masses $m_{v,n}$, factors the feature norm into a within-view and a cross-view consistency C_{inter}, and scores reliability as $\mathcal{R} = \|\mathbf{f}\| N_{\text{eff}}/(N_{\text{eff}}+1)$. We implement it as its paper defines it, without its KNN voting refinement. NormLift defines no epistemic or aleatoric uncertainty. From its quantities we build the reference scores $1/(1+N_{\text{eff}})$, $1-C_{\text{inter}}$ and $1-\mathcal{R}$. *CSS* [12] keeps a Dirichlet posterior over a closed label set, and each pixel votes for the class whose text is closest to its feature. CSS receives the full vocabulary at mapping time, which the open-vocabulary rows do not need. We composite its variance with the linear splat weights of every other channel, not with squared weights as CSS does.

Change baselines. A residual compares an update view with the map rendered at the same pose. We lift every residual to the Gaussians with the splat weights and, as for our score, require it to persist over $k = 2$ observed views. *NCC* is one minus the normalized cross-correlation of observed and rendered grey values in small windows, which ignores a global gain or offset. The *depth residual* is the absolute difference between observed depth and map depth. Its non-oracle row uses the same depth as our re-admission check. Its oracle row uses the RIO10 mesh depth of seqXX_02 and is an upper bound. The *feature residual* is one minus the cosine of rendered and observed features. It is the residual of CL-Splats [16] and the semantic term of MIF's discrepancy score [19], and we re-implement only this per-pixel residual. Four rows use an uncertainty as the change score: the CSS variance, the drop of NormLift's $\mathcal{R}$ during the update, our dispersion $1 - \hat{\rho}$, and a Yager conflict of our own design. For the last one, the CSS class counts of each view form a subjective-logic opinion. Yager's rule [38] fuses the opinions over the views and keeps the conflict mass, which we accumulate. E2-BKI [27] fuses such opinions with Dempster's rule instead. The *fused* row averages the ranks of our score and of the non-oracle depth residual, with equal weights and nothing fitted.

B. Metrics

Segmentation and calibration. We report mIoU over the classes in the ground truth, and pixel accuracy. For the predictive we report ECE [39] with 15 equal-width bins on the maximum probability, NLL, the Brier score, and AURC, the area under the risk-coverage curve with pixels ranked by one minus the maximum probability. A labeled pixel with rendered opacity $\alpha_p < \tau_{\text{cov}}$, or with a tied top class score, counts as wrong. An uncovered pixel also gets the uniform predictive and ranks as the most uncertain pixel for every uncertainty.

Uncertainty ranking. We use the area under the sparsification error [40] in its mean-gap form, $\text{AUSE} = \int_0^1 [\bar{e}_u(f) - \bar{e}_\star(f)]\, df$, on a uniform grid of f. Here $\bar{e}_u(f)$ is the mean error of the pixels left after we remove the fraction f with the highest uncertainty u, and $\bar{e}_\star(f)$ is the same with pixels ordered by their error. View counts produce many tied values of u. Within a tie we use the mean error of the tied pixels, which equals the expectation over a random order. We grade each channel against the error it is meant for. For a field built from M of the 180 mapping frames, a pixel has reducible error ($e_{\text{red}} = 1$) if it is wrong at M frames and right with the full map. It has irreducible error ($e_{\text{irr}} = 1$) if it is wrong in both. All M-frame fields use the full-map geometry, and the targets use the pixels that both maps cover. The epistemic channel should rank e_{red} [31], and the aleatoric channel should rank e_{irr}.

Change. We report average precision (AP) and AUROC for stale content at the Gaussian and at the object level. An object's score is the mean score of its Gaussians. A residual row scores a Gaussian by the largest value of its persistent residual over the update, and the uncertainty rows use their value after the update. A Gaussian that a row never scores ranks last, and we report how many there are. The false-alarm rate (FA) is the share of static Gaussians flagged at least once during the update, and recall (Rec.) is the same share for stale Gaussians. Our threshold targets a nominal per-view false-alarm rate of 10^{-3} (Sec. III-F), so FA grows with the number of update views that see a Gaussian. Every other row gets one threshold, set on the validation scenes with labels so that its per-view false-alarm rate on static Gaussians is 10^{-3}. The baselines thus use labels that our row does not use.

C. Static Scenes: Accuracy, Calibration and Ranking

Table I tests accuracy and the predictive. P0 renders the view-weighted mean of the same features that averaging uses, so the accuracy columns are a parity check. NormLift keeps the averaging feature and differs only in its uncertainty. P2 rescales the P0 logits of each pixel, so it keeps the P0 labels and changes only the confidence. The question is whether P1 or P2 lowers NLL and ECE compared with P0 and with averaging, each at its own temperature. **[TODO: interpret**

TABLE I

SEGMENTATION AND CALIBRATION ON THE REPLICA TEST SCENES (MEAN $\pm$ STD, $N = 6$). MIOU, ACCURACY, ECE, AURC AND AUSE IN %, BRIER $\times 100$, NLL IN NATS. OV: ACCEPTS QUERIES THAT WERE NOT KNOWN AT MAPPING TIME. AUSE_u: AUSE OF THE ROW'S OWN UNCERTAINTY AGAINST THE 0/1 ERROR (THE VARIANCE FOR CSS, u^{ale} FOR OURS). †OUR RE-IMPLEMENTATION FROM THE PAPER. **[TODO: REPLACE PLACEHOLDERS.]**

Method	OV	mIoU↑	Acc.↑	ECE↓	NLL↓	Brier↓	AURC↓	AUSE_u ↓
2D features, per frame	✓	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	–
Averaging [9]†	✓	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	–
NormLift [14]†, $1 - \mathcal{R}$	✓			same predictive as averaging				XX.X ± XX.X
NormLift†, $1 - C_{\text{inter}}$	✓			same predictive as averaging				XX.X ± XX.X
CSS (Dirichlet) [12]†	×	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
Ours P0 (plug-in)	✓	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
Ours P1 (posterior mean)	✓	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
Ours P2 (adaptive β)	✓	as P0		XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X

[TODO: Figure: 2×2 panels. Rows: u^{epi}, u^{ale}. Columns: target e_{red}, target e_{irr}. x-axis: mapping frames $M \in \{10, 30, 90\}$, plus $M = 180$ in the e_{irr} column (e_{red} is empty at $M = 180$). y-axis: AUSE, lower is better. Reference curves: averaging (one minus maximum probability) and, built from NormLift's quantities, $1/(1 + N_{\text{eff}})$, $1 - C_{\text{inter}}$ and $1 - \mathcal{R}$. Replica test, mean $\pm$ std, $N = 6$.]

Fig. 3. Does each uncertainty rank its own error? AUSE of the epistemic and aleatoric channels against reducible error e_{red} and irreducible error e_{irr}, for fields built from M of the 180 mapping frames. If the two channels separate, each is lower on its own target than on the other one. **[TODO: update once measured.]**

once measured. Use the word "calibrated" only if P1 or P2 wins on NLL and ECE.]

Fig. 3 tests whether the two channels separate. Our u^{epi} counts views with a weight that saturates at one full view, so views with a small footprint count fractionally. The score $1/(1+N_{\text{eff}})$, built from NormLift's Kish count of pixel masses, depends only on the relative masses of the views, so ten tiny views count as much as ten full ones. The two can therefore rank e_{red} differently for Gaussians with small footprints. For the aleatoric channel we compare $\hat{\rho}$ (2) with NormLift's cross-view consistency C_{inter}, which, like $\bar{r}$, is biased upward at few views, and with its reliability $\mathcal{R}$, which shrinks with scarce evidence. **[TODO: interpret once measured: is each channel lower on its own target, and is u^{epi} lower than one minus the maximum probability on e_{red}?]**

D. Real Rearrangements: Stale Content

Table II reports stale-content detection. Our score differs from the feature-residual row only in where it takes the cosine: per Gaussian against all of its earlier evidence, not per pixel against the rendered map. Ours also sets its threshold from the mapping pass, resets the Gaussians it flags, and re-admits them after the depth check. The fused row asks whether our score adds to non-oracle depth. In the test scenes, XX.X% of the Gaussians are unobserved and XX.X% are hidden, and they

[TODO: Qualitative figure. Top row: one RIO10 validation frame with a moved or removed object. Bottom row: one Replica frame. Columns: image, query result, u^{epi}, u^{ale}, change score $\bar{c}$ with the flagged Gaussians.]

Fig. 4. Qualitative results. **[TODO: caption once the figure exists.]**

are not in the table. **[TODO: interpret once measured: does ours beat every non-oracle residual in AP, or does it add to the depth residual in the fused row? Report the rows where a baseline wins.] [TODO: If the RIO10 labels map to our vocabulary, add mIoU and ECE after the update on changed and unchanged regions. Otherwise keep the Limitations sentence on synthetic calibration data.]** Fig. 4 shows examples.

E. Ablations

Table III changes one part at a time and keeps the rest. The top block uses Replica. *Pixels instead of views* uses the pixel masses $m_{v,n}$ in place of $\omega_{v,n}$ in the posterior and the support, with $u^{\text{epi}} = 1/(1 + \sum_v m_{v,n})$. Its $\hat{\rho}$ still combines view means, now with pixel-mass weights, since a pixel-level Q would need an extra render of squared weights. The agreement rows replace $\hat{\rho}$ (2) by the uncorrected $\bar{r}$, by the prior-shrunk $\|\mathbf{R}\|/(W_1 + c_0)$, or by the shrunk mean length $\tilde{r} = \bar{r} - (1-\bar{r})/(W_1+1)$. Note that $\tilde{r} = 1$ whenever $\bar{r} = 1$, so $\tilde{r}$ does not correct the one-view case. *Centered features* subtract the scene-mean feature before the dispersion and the change test. The full method uses **[TODO: the predictive chosen on validation (P0, P1 or P2, compared in Table I)]**. The bottom block uses RIO10. The persistence and bucket rows vary k and the retention λ of the change test. The threshold rows replace the history-conditioned null by one pooled null per scene, by a null of per-Gaussian maxima over the mapping pass, or by one fixed threshold from the validation scenes. *Re-admission without depth* re-admits a Gaussian after the views alone. The *pixel-level variant* turns every part off. It counts

TABLE II

STALE-CONTENT DETECTION ON THE RIO10 TEST SCENES (MEAN ± STD, $N = 7$). AP, AUROC, FA AND REC. IN %. INPUT: I = IMAGE, D = OBSERVED DEPTH, F = FEATURES, V = VOCABULARY AT MAPPING TIME, M = MESH DEPTH OF THE NEW RESCAN. †OUR RE-IMPLEMENTATION (FOR [16], [19], OF THE PER-PIXEL FEATURE RESIDUAL ONLY). THE ORACLE ROW READS THE NEW MESH, WHICH A ROBOT DOES NOT HAVE. **[TODO: REPLACE PLACEHOLDERS. RIO10 ACCESS PENDING.]**

		Gaussian level		Object level		Operating point	
Method	Input	AP↑	AUROC↑	AP↑	AUROC↑	FA↓	Rec.↑
NCC photometric residual	I	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
Depth residual	D	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
Feature residual, $k = 2$ [16], [19]†	F	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
CSS variance [12]†	F+V	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
Yager conflict (our design)	F+V	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
NormLift reliability drop [14]†	F	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
Dispersion $1 - \hat{\rho}$ (ours)	F	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
Ours: test, reset, re-admission	F+D	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
Ours fused with depth residual	F+D	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X
Depth residual, oracle (upper bound)	M	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X	XX.X ± XX.X

TABLE III

ABLATIONS ON THE TEST SCENES, MEAN OVER $N = 6$ (TOP) AND $N = 7$ (BOTTOM). EACH ROW CHANGES ONE PART OF THE FULL METHOD. AUSE: u^{epi} AGAINST e_{red} (EPI) AND u^{ale} AGAINST e_{irr} (ALE), AVERAGED OVER THE 10-, 30- AND 90-FRAME FIELDS. AP AT THE GAUSSIAN (G) AND OBJECT (O) LEVEL. KEPT: SHARE OF STALE GAUSSIANS IN THE ROBUST SET AFTER THE UPDATE. ALL METRICS EXCEPT NLL IN %. **[TODO: REPLACE PLACEHOLDERS.]**

Replica test				AUSE↓	
Variant	mIoU↑	ECE↓	NLL↓	epi	ale
Full method	XX.X	XX.X	XX.X	XX.X	XX.X
Pixels instead of views	XX.X	XX.X	XX.X	XX.X	XX.X
Uncorrected mean length $\bar{r}$	XX.X	XX.X	XX.X	XX.X	XX.X
Prior-shrunk agreement	XX.X	XX.X	XX.X	XX.X	XX.X
Shrunk mean length $\tilde{r}$	XX.X	XX.X	XX.X	XX.X	XX.X
Centered features	XX.X	XX.X	XX.X	XX.X	XX.X
Pixel-level variant	XX.X	XX.X	XX.X	XX.X	XX.X

RIO10 test	AP↑				
Variant	G	O	FA↓	Rec.↑	Kept↓
Full method	XX.X	XX.X	XX.X	XX.X	XX.X
No persistence ($k = 1$)	XX.X	XX.X	XX.X	XX.X	XX.X
Persistence $k = X$	XX.X	XX.X	XX.X	XX.X	XX.X
Two buckets, $\lambda = XX.X$	XX.X	XX.X	XX.X	XX.X	XX.X
Pooled null	XX.X	XX.X	XX.X	XX.X	XX.X
Per-Gaussian max null	XX.X	XX.X	XX.X	XX.X	XX.X
Fixed threshold (val.)	XX.X	XX.X	XX.X	XX.X	XX.X
Re-admission without depth	XX.X	XX.X	XX.X	XX.X	XX.X
Centered features	XX.X	XX.X	XX.X	XX.X	XX.X
Pixel-level variant	XX.X	XX.X	XX.X	XX.X	XX.X

pixels, uses $\tilde{r}$, has no change test, forgets by decaying the evidence of Gaussians with low $\bar{r}$, and scores change by $1 - \bar{r}$. We keep a part only if removing it hurts on the validation scenes. **[TODO: interpret once measured, then rename the method and adjust the claims to the parts that remain.]**

Run time and memory. We time one validation scene per dataset on one NVIDIA GeForce RTX 5070 Ti with an Intel Core i5-14600KF and no other job running. On Replica (RIO10), a keyframe takes XX.X (XX.X) ms in total: XX.X (XX.X) ms for LSeg feature extraction, XX.X (XX.X) ms for the render and backward pass, and XX.X (XX.X) ms for the statistics, change test and re-admission. Peak GPU memory is XX.X GB. Averaging and NormLift reuse the sums of our backward pass and add no render or backward pass. CSS needs its own backward pass for the class votes, which adds XX.X ms. Fitting the 3D-GS map takes XX.X min per scene, and the field state takes XX.X bytes per Gaussian.

F. Limitations

The geometry is fixed after mapping, so moved content is not inserted at its new place. The features limit the absolute accuracy. Every row uses LSeg, and fusion cannot fix a class that the features confuse in every view. The change statistic is a known residual, and its threshold assumes a static mapping sequence. Content that moves during mapping enters the null and raises the threshold. The mapping-pass scores are also in-sample, because the map and the PCA are fit on those frames. The false-alarm rate on a new sequence can thus exceed the nominal rate. The 5 cm depth check cannot keep out the Gaussians of a thin removed object, such as a book on a table, since they lie within 5 cm of the surface behind them. We re-implemented averaging, NormLift, CSS and the feature residual of CL-Splats from their papers, and differences from the authors' code may remain. **[TODO: Keep only if the RIO10 class map fails:]** The predictive is evaluated on synthetic data (Replica) only.

V. CONCLUSION

We presented an open-vocabulary semantic field for 3D Gaussian Splatting that counts evidence in views and keeps little evidence and conflicting evidence apart. A known feature residual, with a threshold from the map's own mapping pass, drives the reset and the depth-checked re-admission of changed Gaussians. **[TODO: one sentence with the main measured result on RIO10 and one on Replica.]** Future work should insert new geometry where content has moved and test the field with other feature extractors.

REFERENCES

[1] S. Lee, H. Yu, G. Kim, and S. Choi, "LAMP: Implicit language map for robot navigation," *IEEE Robot. Autom. Lett.*, vol. 10, no. 12, pp. 12 365–12 372, 2025.

[2] K. Nagami *et al.*, "VISTA: Open-vocabulary, task-relevant robot exploration with online semantic Gaussian splatting," *IEEE Robot. Autom. Lett.*, vol. 11, no. 3, pp. 3150–3157, 2026.

[3] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian splatting for real-time radiance field rendering," *ACM Trans. Graph.*, vol. 42, no. 4, pp. 139:1–139:14, 2023.

[4] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 139, 2021, pp. 8748–8763.

[5] M. Qin, W. Li, J. Zhou, H. Wang, and H. Pfister, "LangSplat: 3D language Gaussian splatting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 20 051–20 060.

[6] S. Zhou *et al.*, "Feature 3DGS: Supercharging 3D Gaussian splatting to enable distilled feature fields," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 21 676–21 685.

[7] Y. Wu *et al.*, "OpenGaussian: Towards point-level 3D Gaussian-based open vocabulary understanding," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, 2024, pp. 19 114–19 138.

[8] J.-C. Shi, M. Wang, H.-B. Duan, and S.-H. Guan, "Language embedded 3D Gaussians for open-vocabulary scene understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 5333–5343.

[9] J. Joseph, B. Amrutur, and S. Bhatnagar, "Gradient-weighted feature back-projection: A fast alternative to feature distillation in 3D Gaussian splatting," in *Proc. SIGGRAPH Asia Conf. Papers*, 2025, pp. 1–12.

[10] W. Vega-Brown, M. Doniec, and N. Roy, "Nonparametric Bayesian inference on multivariate exponential families," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 27, 2014.

[11] L. Gan, R. Zhang, J. W. Grizzle, R. M. Eustice, and M. Ghaffari, "Bayesian spatial kernel smoothing for scalable dense semantic mapping," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 790–797, 2020.

[12] J. Wilson *et al.*, "Modeling uncertainty in 3D Gaussian splatting through continuous semantic splatting," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2025, pp. 3284–3290.

[13] ——, "LatentBKI: Open-dictionary continuous mapping in visual-language latent spaces with quantifiable uncertainty," *IEEE Robot. Autom. Lett.*, vol. 10, no. 4, pp. 3102–3109, 2025.

[14] Y. Zang, D. Li, D. Engel, S. Park, and I. Viola, "NormLift: From lifted features to semantic reliability in 3D Gaussian splatting," arXiv:2609.18898, 2026.

[15] Z. Lu, J. Ye, and J. Leonard, "3DGS-CD: 3D Gaussian splatting-based change detection for physical object rearrangement," *IEEE Robot. Autom. Lett.*, vol. 10, no. 3, pp. 2662–2669, 2025.

[16] J. Ackermann *et al.*, "CL-Splats: Continual learning of Gaussian splatting with local optimization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 7808–7817.

[17] L. Zeng *et al.*, "GaussianUpdate: Continual 3D Gaussian splatting update for changing environments," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 25 800–25 809.

[18] Y. He, J. Yu, G. Chen, and H. Zhang, "VG-Mapping: Variation-aware density control for online 3D Gaussian mapping in semi-static scenes," *IEEE Robot. Autom. Lett.*, vol. 11, no. 10, pp. 11 561–11 568, 2026.

[19] P. Jiang, H. Liu, J. Jin, W. Wang, and X. Li, "Learning to evolve: Multimodal interactive fields for robust humanoid navigation in dynamic environments," in *Proc. Robot.: Sci. Syst. (RSS)*, 2026.

[20] K. V. Mardia and P. E. Jupp, *Directional Statistics*. Chichester, U.K.: Wiley, 2000.

[21] J. Straub *et al.*, "The Replica dataset: A digital replica of indoor spaces," arXiv:1906.05797, 2019.

[22] S. Zhi, T. Laidlow, S. Leutenegger, and A. J. Davison, "In-place scene labelling and understanding with implicit scene representation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 15 838–15 847.

[23] J. Wald, T. Sattler, S. Golodetz, T. Cavallari, and F. Tombari, "Beyond controlled environments: 3D camera re-localization in changing indoor scenes," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 467–487.

[24] J. Wald, A. Avetisyan, N. Navab, F. Tombari, and M. Nießner, "RIO: 3D object instance re-localization in changing indoor environments," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 7658–7667.

[25] J. Cheng, J.-N. Zaech, L. Van Gool, and D. P. Paudel, "Occam's LGS: An efficient approach for language Gaussian splatting," in *Proc. Brit. Mach. Vis. Conf. (BMVC)*, 2025.

[26] L. Kish, *Survey Sampling*. New York: Wiley, 1965.

[27] J. Kim, M. Jeon, J. Min, K. Kwak, and J. Seo, "E2-BKI: Evidential ellipsoidal Bayesian kernel inference for uncertainty-aware Gaussian semantic mapping," *IEEE Robot. Autom. Lett.*, vol. 11, no. 3, pp. 2378–2385, 2026.

[28] A. Sheppard *et al.*, "SLIM-VDB: A real-time 3D probabilistic semantic mapping framework," *IEEE Robot. Autom. Lett.*, vol. 11, no. 3, pp. 2386–2393, 2026.

[29] D. Maggio and L. Carlone, "Bayesian fields: Task-driven open-set semantic Gaussian splatting," arXiv:2503.05949, 2025.

[30] S. Chopra, J. Liang, G. Seneviratne, and D. Manocha, "PhysGS: Bayesian-inferred Gaussian splatting for physical property estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 18 980–18 990.

[31] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?" in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017.

[32] K. V. Mardia and S. A. M. El-Atoum, "Bayesian inference for the von Mises-Fisher distribution," *Biometrika*, vol. 63, no. 1, pp. 203–206, 1976.

[33] A. Banerjee, I. S. Dhillon, J. Ghosh, and S. Sra, "Clustering on the unit hypersphere using von Mises-Fisher distributions," *J. Mach. Learn. Res.*, vol. 6, no. 46, pp. 1345–1382, 2005.

[34] B.-T.-H. Chu and G.-W. Kim, "Semantic-augmented 3D Gaussian splatting for visual localization in complex indoor environments," *IEEE Robot. Autom. Lett.*, vol. 11, no. 3, pp. 3852–3859, 2026.

[35] B. Li, K. Q. Weinberger, S. Belongie, V. Koltun, and R. Ranftl, "Language-driven semantic segmentation," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[36] V. Ye *et al.*, "gsplat: An open-source library for Gaussian splatting," *J. Mach. Learn. Res.*, vol. 26, no. 34, pp. 1–17, 2025.

[37] B. Glocker, S. Izadi, J. Shotton, and A. Criminisi, "Real-time RGB-D camera relocalization," in *Proc. IEEE Int. Symp. Mixed Augmented Reality (ISMAR)*, 2013, pp. 173–179.

[38] R. R. Yager, "On the Dempster-Shafer framework and new combination rules," *Information Sciences*, vol. 41, no. 2, pp. 93–137, 1987.

[39] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 70, 2017, pp. 1321–1330.

[40] E. Ilg *et al.*, "Uncertainty estimates and multi-hypotheses networks for optical flow," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 677–693.